

EXPLORATORY DATA ANALYSIS OF SAAS PRODUCT USAGE & CUSTOMER RETENTION

Subscription Behaviour, Product
Usage & Customer Retention



TOOLS USED

Python, Pandas, Matplotlib, Seaborn & Plotly

PRESENTED BY

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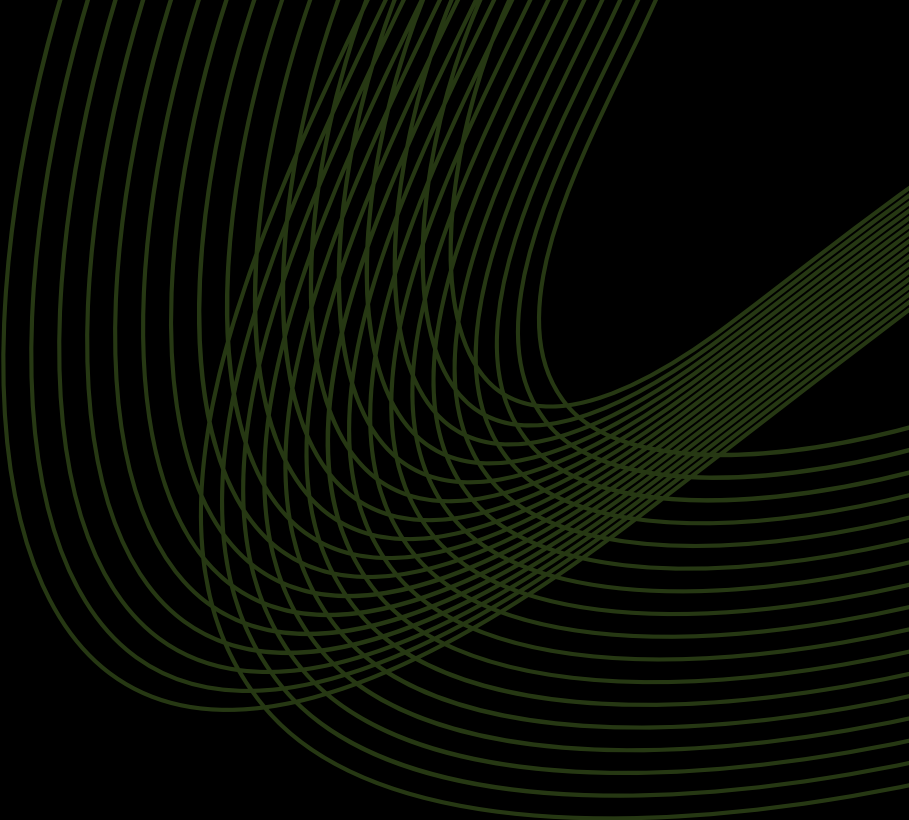


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Problem Statement:

Modern SaaS businesses operate across diverse customer segments with varying usage and retention behaviors, making it challenging to pinpoint the engagement and operational factors that influence churn and long-term revenue.

Objective:

To explore the SaaS customer usage and retention dataset to uncover patterns, insights, and trends that can help businesses make data-driven decisions around customer engagement, retention, and revenue growth.



EDA WORKFLOW

For this analysis, a structured workflow was followed, involving data collection, understanding, cleaning, exploration, and summarization of insights to allow a clear understanding of the dataset and its trends.

01

Data Collection

- Gathered the Google Play Store dataset from Kaggle

02

Data Understanding & Anomaly Detection

- Looked at data distributions
- Found missing values, outliers, and unusual patterns

03

Data Cleaning & Treatment

- Fixed missing or incorrect values
- Standardized formats for consistency

04

Exploratory Analysis

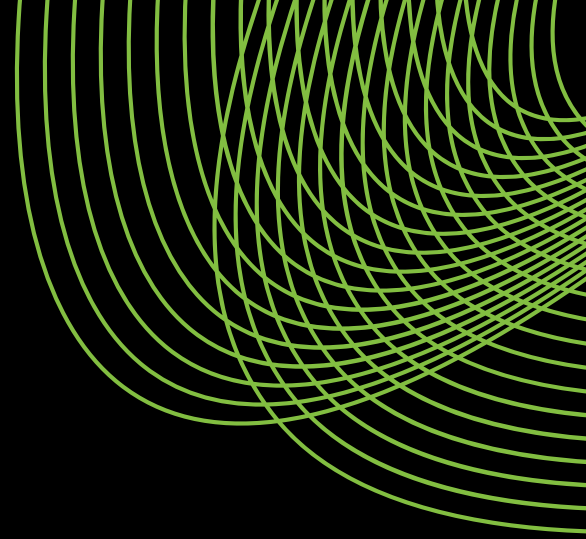
- Univariate Analysis: e.g Customer Segment, Complaint Type, Contract Type etc
- Bivariate Monthly Logins vs Features Used
- Multivariate Analysis: CSAT vs Contract Type vs Customer Segment

05

Insights & Reporting

- Summarized patterns and trends
- Highlighted key findings for developers and businesses

KEY QUESTIONS EXPLORED



- How do user demographics (age and gender) influence shopping behavior and engagement patterns on the platform?
- Which stages of the shopping journey (browsing, cart, checkout) contribute most to drop-offs and missed conversions?
- Do occasional shoppers present a greater growth opportunity compared to already frequent buyers?
- How effective are product recommendations in influencing purchase decisions, and where do they fall short?
- What role does pricing perception play in cart abandonment and cross-platform comparison behavior?
- How do customer support responsiveness and delivery reliability impact overall satisfaction and trust?
- How do reviews and post-purchase experiences influence future purchase intent and platform credibility?
- Which satisfaction drivers most strongly contribute to repeat purchases and long-term loyalty?



DATA OVERVIEW

The dataset provides insights into SaaS customer behavior, covering demographics, subscription characteristics, product usage, billing activity & support interactions. It also captures customer engagement signals, satisfaction metrics, and feedback indicators that are relevant for understanding retention and churn dynamics.

Data Source: Kaggle

Dataset Size

10000

Records

32

Features

Purchase Diversity

3

Customer Segments

7

Countries

3

Contract Types



DATA OVERVIEW

Below is a detailed description of the feature set:

Dataset Features	Type	Feature Description
customer_id	Date	Unique identifier assigned to each customer
gender	Categorical	Gender of the customer
age	Numerical	Age of the customer in years
country	Categorical	Country where the customer is located
customer_segment	Categorical	Business-defined customer classification (Individual, SMB, Enterprise)
tenure_months	Numerical (Discrete)	Number of months the customer has been subscribed to the service
signup_channel	Categorical	Channel through which the customer signed up (Organic, Ads, Referral)
contract_type	Categorical	Type of subscription contract (Monthly, Quarterly, Annual)
monthly_logins	Numerical (Discrete)	Total number of platform logins by the customer in a month
weekly_active_days	Numerical (Discrete)	Average number of days per week the customer actively used the platform
avg_session_time	Numerical (Continuous)	Average duration of a customer session on the platform
features_used	Numerical (Discrete)	Number of distinct product features used by the customer
usage_growth_rate	Numerical (Continuous)	Percentage change in usage over a recent time period
last_login_days_ago	Numerical (Discrete)	Number of days since the customer last logged into the platform
monthly_fee	Numerical (Discrete)	Monthly subscription fee charged to the customer
total_revenue	Numerical (Discrete)	Total revenue generated from the customer over their lifetime

DATA OVERVIEW

Continued..

Dataset Features	Type	Feature Description
city	Categorical	City where the customer resides
payment_method	Categorical	Payment method used by the customer
payment_failures	Numerical (Discrete)	Number of failed payment attempts recorded
discount_applied	Categorical	Indicates whether a discount was applied to the subscription
price_increase_last_3m	Categorical	Indicates whether the customer experienced a price increase in the last three months
support_tickets	Numerical (Discrete)	Number of customer support tickets raised
avg_resolution_time	Numerical (Continuous)	Average time taken to resolve customer support tickets
complaint_type	Categorical	Category of complaint raised by the customer
csat_score	Numerical (Continuous)	Customer Satisfaction Score
escalations	Numerical (Discrete)	Number of support cases escalated beyond first-level support
email_open_rate	Numerical (Continuous)	Percentage of marketing or communication emails opened
marketing_click_rate	Numerical (Continuous)	Percentage of marketing emails where links were clicked
nps_score	Numerical (Discrete)	Net Promoter Score provided by the customer
survey_response	Categorical	Indicates whether the customer responded to surveys
referral_count	Numerical (Discrete)	Number of new customers referred by the existing customer
churn	Numerical (Binary)	Indicates whether the customer discontinued the service (0 = No, 1 = Yes)

DATA QUALITY CHALLENGES & ANOMALIES

Few inconsistencies were found in the dataset, which could have affected the analysis if left unaddressed.

DATA ANOMALIES

- nps_score is stored as an int64. While numeric, it functions as an ordinal / categorical business metric (e.g., Detractors, Passives, Promoters). Depending on analytical objectives, it may require grouping or categorical transformation rather than being treated as a continuous variable.
- Several categorical attributes are stored as object (string) types, including: gender, country, city, customer_segment, signup_channel, contract_type, payment_method, complaint_type, discount_applied, price_increase_last_3m, and survey_response. Converting these fields to the category data type would improve analysis without altering semantic meaning.
- complaint_type contains 2,045 missing values (20.45%)
- No duplicate or conflicting columns were observed in the final dataset.



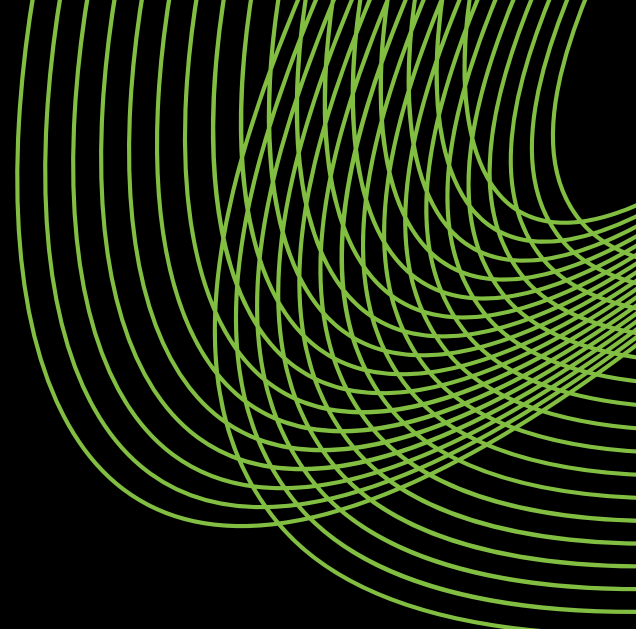
DATA CLEANING & TREATMENT

Inconsistent and missing values were addressed, and key features were cleaned and standardized for analysis.

DATA CLEANING SUMMARY

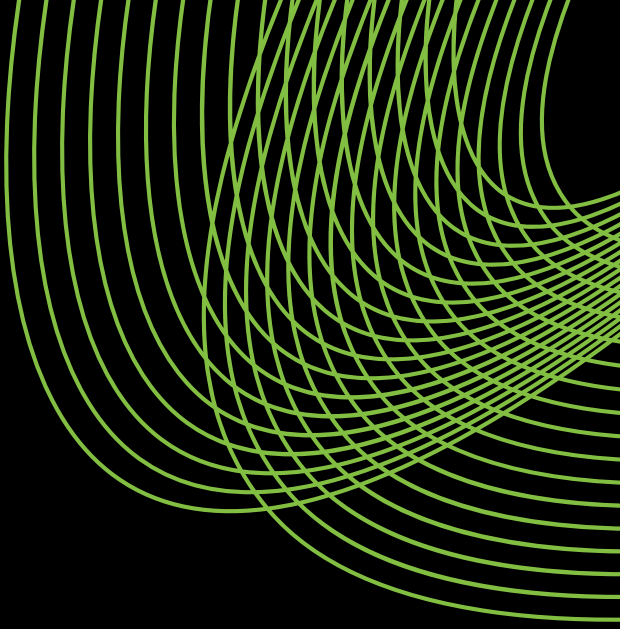
- Several categorical features, including gender, country, city, customer_segment, signup_channel, contract_type, payment_method, complaint_type, discount_applied, price_increase_last_3m, and survey_response, were standardized through data type conversion to ensure consistency and improved analytical performance.
- The **complaint_type** column contained missing values representing customers who did not raise any complaints. Rather than imputing these missing values with the mode (which would introduce bias), missing entries were replaced with a meaningful category such as "No Complaint".





INSIGHTS





Customer Profile & Account Overview

